AUTOMATING EVIDENCE SYNTHESIS WITH MULTI-AGENT LLMS

SAM: Systematic Agent for Meta-analysis

A multi-agent LLM system for automating systematic reviews and meta-analyses

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01 Introduction & objective

Systematic reviews and meta-analyses are the **gold standard of evidence synthesis**, yet they are slow and expensive: on average `12.5 months` per review (Borah et al., 2017), and large reviews can exceed `$300,000`.

Our objective is not merely to test a pipeline, but to build an **open, free tool** that saves thousands of hours and euros while delivering results **at least as good as human reviewers**.

We built a **9-agent system** that mirrors the standard meta-analysis workflow, and benchmarked it against human expert decisions using two language models: an open model (`Qwen 2.5 7B`) and a frontier model (`Claude Opus`).

02 Methodology

Validated against a **real, published gold standard**: a meta-analysis on physical-activity promotion (`Collazo-Castiñeira et al., 2025`), with full lists of included / excluded studies and exclusion reasons.

- Each agent **reproduces one stage** of the standard meta-analysis pipeline.
- Agreement with human expert decisions is measured with **Cohen's κ**, sensitivity and specificity.
- **Matched instructions** across both models for a fair open-vs-frontier comparison.

Cohen's κ measures agreement beyond chance: **1 = perfect, 0 = chance-level**. For ordinal risk-of-bias ratings we use **quadratic κ**, which penalizes large disagreements more than small ones.

03 The pipeline · A1–A9

- A1 **Protocol** — defines PICO inclusion criteria
- A2 **Search** — queries scientific databases
- **A3 Title / abstract screening**
- **A4 Full-text screening**
- A5 **Data extraction**
- **A6 Risk-of-bias assessment**
- A7 **Synthesis** — meta-analysis
- A8 **Manuscript drafting**
- A9 **Explainability**

Stages evaluated in this study: A3, A4, A6

04 Explainability

A9 · EVERY DECISION IS TRACEABLE

SAM is not a black box. Every decision is logged with its justification in an explainability report and a paper-by-paper spreadsheet (agent A9), so each inclusion / exclusion is **fully traceable**.

EXAMPLE DECISION LOG · A4 FULL-TEXT

PAPER	"Effect of a digital intervention on physical activity in institutionalized older adults…"
DECISION	EXCLUDE
REASON	Population outside PICO criteria — does not match the protocol's target age range.

06 PRISMA flow

- **Records screened** `20,750`
- **Abstract screening** → Excluded at abstract `19,198`
- **Relevant · full-text assessed** `1,552` → Excluded at full-text (with reason) `1,486`
- **Studies included** `66`

Human gold-standard funnel · Collazo-Castiñeira et al., 2025

· Models benchmarked

FRONTIER **Claude Opus**
Proprietary frontier model. Strongest agreement at full-text screening (κ = 0.62).

OPEN **Qwen 2.5 7B**
Open-weights, 7B parameters — runs locally and for free. Reliable at abstract screening only.

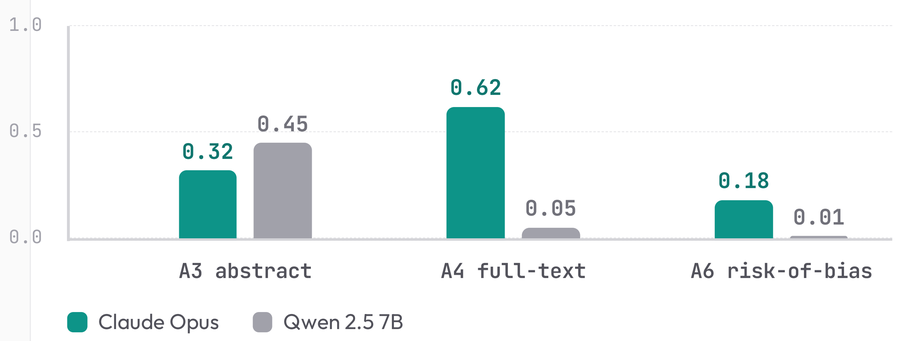
Both models ran under **identical, matched instructions**, isolating model capability as the only variable in the open-vs-frontier comparison.

05 Results — agreement with the human gold standard

Stage / metric	Human gold	Claude Opus frontier	Qwen 2.5 7B open
A3 · ABSTRACT SCREENING (N = 300, matched instructions)			
Cohen's κ	—	0.32	0.45
Included	—	52	77
Excluded	—	226	154
A4 · FULL-TEXT SCREENING (N ≈ 386)			
Cohen's κ	—	**0.62***	0.05
Sensitivity	—	0.78	0.76
Specificity	—	0.93	0.78
A6 · RISK OF BIAS (N = 61, quadratic κ^2)			
Quadratic κ^2	—	0.18	0.01
RoB · Low / Some / High	**3 / 37 / 21**	3 / 27 / 31	4 / 57 / 0

* Substantial agreement — see takeaway. · Human "gold" is the reference standard; counts shown where comparable. · Single run per condition.

Agreement (κ) by evaluated stage
Claude Opus vs Qwen 2.5 7B



KEY TAKEAWAY

At full-text screening, SAM with a frontier model reaches **κ = 0.62** — **substantial agreement** with the human expert, comparable to typical human inter-reviewer agreement in screening tasks.

The **open model** reaches reliable agreement at **abstract screening** but not at full text.

07 · CONCLUSION

Accelerating evidence synthesis without giving up expert judgment

SAM **automates the full systematic-review cycle** and reproduces expert decisions with **substantial agreement**, while keeping every step **traceable**. An open, free tool to accelerate evidence synthesis without giving up expert-level judgment.

08 · LIMITATIONS & FUTURE WORK

Single-domain validation (physical activity). **Single run per condition** — no significance testing yet. Statistical synthesis (A7) and **multi-annotator validation** are future work.

Have you run a meta-analysis and want to contribute it as a gold standard? **Write to us** — let's accumulate more evidence together. `iqueipo.pg24@gmail.com` · `iqueipopg@alu.comillas.edu`

REFERENCES

Borah, R., Brown, A. W., Capers, P. L., & Kaiser, K. A. (2017). Analysis of the time and workers needed to conduct systematic reviews of medical interventions using data from the PROSPERO registry. *BMJ Open, 7*(2), e012545.
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